

7-7 Equations and Inequalities Problem Solving

Equation Detective Agency | Standard: 6.EE.7 | Color-by-Number Practice

Learning Goal: I can model and solve real-world problems using equations and inequalities.

Directions: Solve each problem. Match each answer to the Color Code. On the next page, color every box with that problem number using the matching color.

Vocabulary: model, equation, inequality, solution

Explain one: I solved problem ___ by ___. Because ___, my answer is ___.

Solve First

1. A game costs \$7. You have \$42. How many games can you buy?

Work: _____
Answer: _____

2. You need at least 30 points and have 18. How many more points do you need?

Work: _____
Answer: _____

3. A box holds 8 books. How many boxes for 56 books?

Work: _____
Answer: _____

4. You spent \$15 from \$50. How much is left?

Work: _____
Answer: _____

5. Each ticket costs \$6. How much for 9 tickets?

Work: _____
Answer: _____

6. You need more than 25 stickers and have 11. How many more to reach 26?

Work: _____
Answer: _____

7. A ride lasts 4 minutes. How many minutes for 8 rides?

Work: _____
Answer: _____

8. A student read 12 pages each day for 5 days. How many pages?

Work: _____
Answer: _____

9. There are 72 chairs in rows of 9. How many rows?

Work: _____
Answer: _____

10. A score increased by 14 to become 41. What was the start?

Work: _____
Answer: _____

11. Each team has 6 players. How many teams from 48 players?

Work: _____
Answer: _____

12. You need at least 100 cans and have 67. How many more?

Work: _____
Answer: _____

Quick check: Before coloring, make sure every answer appears in the Color Code on the next page.

7-7 Mystery Picture: Detective Badge Mosaic

Solve a problem, match the answer to a color, then color every box with that problem number.

Color Code	Color Code	Color Code	Color Code
6 = red	12 = orange	7 = yellow	35 = green
54 = blue	15 = purple	32 = pink	60 = brown
8 = gray	27 = teal	33 = black	

2	2	1	2	2	2	3	3	3	7	4	6
7	5	5	6	5	5	6	6	6	7	9	12
6	5	8	9	8	8	9	9	9	12	9	6
3	6	10	2	2	4	2	2	4	2	11	6
3	6	9	2	5	5	5	7	4	2	9	6
3	6	9	1	5	8	11	7	4	12	9	6
5	8	11	4	6	12	1	10	4	2	11	8
5	8	11	3	8	8	10	6	5	2	11	8
5	8	11	5	5	5	4	5	1	3	11	8
7	10	6	4	4	2	3	3	4	1	2	10
7	12	12	1	11	11	12	12	12	2	2	11
9	4	9	8	8	8	9	3	9	10	11	11

Early finisher writing: Choose one problem and explain your thinking in one complete sentence: Because ___, I ___.

Teacher Answer Key

#	Answer	Color	Teacher Tip
1	6	red	Write an equation or inequality first.
2	12	orange	Write an equation or inequality first.
3	7	yellow	Write an equation or inequality first.
4	35	green	Write an equation or inequality first.
5	54	blue	Write an equation or inequality first.
6	15	purple	Write an equation or inequality first.
7	32	pink	Write an equation or inequality first.
8	60	brown	Write an equation or inequality first.
9	8	gray	Write an equation or inequality first.
10	27	teal	Write an equation or inequality first.
11	8	gray	Write an equation or inequality first.
12	33	black	Write an equation or inequality first.

Use note: This file is original practice aligned to the lesson skill. It is designed for low-prep review, centers, sub plans, or early finishers.